PROTECTING SPACE ASSETS: A LEGAL ANALYSIS OF "KEEP-OUT ZONES"

*F. Kenneth Schwetje**

Aerospace Daily recently reported that the Strategic Defense Initiative Organization (SDIO) made reference in a Senate report to "Keep-Out Zones" as a useful way of protecting space-based objects.[1] The notion of insulating space objects with protective zones is neither novel nor original with the SDIO. This paper will demonstrate that the concept has been discussed by a number of authors since the earliest days of space exploration.[2] While the topic has been of interest academically, no genuine need to establish any such zones has yet emerged. The renewed interest in "Keep-Out Zones," "Safety Zones," "Zones of Exclusivity," and a host of other names has been promoted by a growing concern that any space-based weapons systems deployed by the United States could be easily neutralized by Soviet "space mines."[3] There seems to be some questions, at least in the West, as to whether or not such zones can be established under the current outer space legal regime. What follows first is a discussion of the technical concept of these keep-out zones. Next, various analogous situations on Earth are examined in an effort to determine how international law treats the exercise of such a claim, falling short of sovereignty.

Soviet jurists have far less difficulty with the notion than we do in the United States and Europe. An examination of the writings of preeminent Soviet scholars demonstrates that they have already expressed a willingness to establish zones around space objects, a fact often

---

* Chief, Air and Space Law, J.A.G.'s Dept., U.S.A.F., Washington, D.C.; Chair, A.B.A. Aerospace Law Committee.

The views and conclusions expressed in this paper are those of the author and do not necessarily reflect the official policy or position of the Department of Defense, United States Air Force, or the United States Government.

1. *SDIO Response to Senate Report Mentions 'Keep-Out Zones,'* AEROSPACE DAILY, April 15, 1987, p. 85.

2. MYRES S. MCDOUGAL, HAROLD LASSWELL AND IVAN VLASIC, LAW AND PUBLIC ORDER IN SPACE 301-311 (1963).

3. These are satellites deployed within lethal range of a target space object. In the optimum mode of deployment, they would continuously trail their target. Using conventional or nuclear charges, the space mine could destroy its target on command from Earth or could be salvage fused to explode when attacked or disturbed.

overlooked by Western commentators. In contrast to the consistent Soviet position, the comments of some Western writers are set forth. The paper concludes with recommendations on how the concept might be employed.

The Notion of Zones

Various proposals have been made on how a zone regime might be fashioned. Some are predicated on bilateral or multilateral agreements, others could be established unilaterally. Often, the same terminology is used to describe similar, but not identical, arrangements for separating space objects.

Frequently, the idea of zones is part of a more elaborate scheme to devise "rules of the road" for outer space traffic.[4] One author includes "Keep-Out Zones" as a cardinal rule of the road. He defines the zone as "an area of space through which the space objects of other nations could not pass without permission."[5] The model Keep-Out Zone envisioned by this author and that of the Office of Technology Assessment might include the following elements:[6]

- Keep 100 kilometers and three degress out-of-plane from foreign satellites below 5000 km.
- Keep 500 kilometers from foreign satellites above 5000 kilometers except those within 500 kilometers of geosynchronous altitude.
- One pre-announced close approach at a time is allowed.
- In the event of a violation of the rules above, the nation of registry of the satellite that most recently initiated a maneuver "burn" is at fault and guilty of trespass.
- Satellites trespassing upon keep-out zones may be forcibly prevented from continued trespass.

In the late 1970s, *M. I. Lazarev* wrote an article advocating the use of protective zones in space.[7] He considered it obvious that security zones would be created around "space cities." The dimensions of the zone could be determined by technical specialists. Peaceful passage would be permitted through the zones. *Lazarev* opined that absent security zones,

4. Collins & Williams, *Towards Traffic Systems for Near-Earth Space*, PROC. 29TH COLLOQ. L. OUTER SPACE 161 (1986).

5. DalBello, *'Rules of the Road': Legal Measures to Strengthen the Peaceful Uses of Outer Space*, PROC. 28TH COLLOQ. L. OUTER SPACE 9 (1985).

6. U.S. CONG. OFF. TECH. ASSESS., ANTI-SATELLITE WEAPONS, COUNTERMEASURES, AND ARMS CONTROL at 136 (1985).

7. Lazarev, *Future Space Cities*, 5 ANNALS AIR & SPACE L. 529 (1980).

it would be necessary to establish the space-equivalent of the territorial sea.[8]

Another similar idea is the product of legal research done by *Lt. Col. A. J. Butler*, USAF (ret.). *Colonel Butler* designed a system to protect manned space assets. Under Butler's scheme, two distinct types of zones would be established. The first would be a true safety zone that required permission prior to entry and would be of limited dimension. The extent of the safety zone would only be as large as necessary to protect activities in which safety was a prime factor. Outside the safety zone, a Space Defense Identification Zone (SPADIZ) would be established. The SPADIZ would be quite large; all approaching spacecraft would be required to identify themselves and give prior notice of approach.[9]

Blau and *Goure* proposed the creation of "exclusionary zones," the penetration of which provides a recognizable and acceptable presumption of a threat.[10] They define the crossing of such zones as creating a reasonable expectation of attack, and they would, accordingly, preserve the right to respond with force.

The most elaborate system has been proposed by *Albert Wohlstetter* and *Brian Chow*.[11] However, the "Space-Defense Zones" (SDZ) differ in three aspects from other "Keep-Out Zone" proposals. First, rather than attempting to protect every satellite with a critical mission, an SDZ protects the mission itself. Secondly, the zones are not attached to the satellites; *Wohlstetter* and *Chow* reserve entire sectors for satellites of various defense groupings. And, finally, they find previous proposals silent or ineffective in dealing with violations.

They have fashioned a number of self-defense zones to be assigned to the Western Alliance or the Soviet Bloc. Each side would have the right to decide which of the other side's satellites may remain in its zone. In the event of a violation, there is an explicit right to inspect, expel or render harmless invading satellites. This SDZ arrangement would make the entire "shell" inaccessible to an attacker, so the threat of space mines would be lessened.

---

8. *Id.*, at 532.

9. To the author's knowledge, the concept of SPADIZ originated with Lt. Col. Butler and is contained in unpublished memoranda written by him.

10. T. BLAU & D. GOURE, MILITARY AND DIPLOMATIC ISSUES IN ACTIVE SPACE DEFENSE 39 (1980).

11. A. WOHLSTETTER & B. CHOW, SELF DEFENSE ZONES IN SPACE 29-32 (1986).

Zones On and Above Earth

In the earthly environment, the international community has generally not objected to the establishment of special zones if the zones under the circumstances were reasonable and did not unduly hamper or interfere with another nation's freedom to navigate the seas or supra-adjacent airspace. Important in the following discussion is the notion that the key to gaining international acceptance of such zones has been that they extend no claim of sovereignty. There is a clear distinction between sovereignty and the right to exercise a preventive, protective, or regulatory jurisdiction.[12]

Various zones have been declared and legally recognized. These zones have various names and serve diverse purposes.[13] During periods of hostility, extensive controlled areas have been established. During World War I, thiry-three defensive zones were created by the United States and in World War II, seventeen maritime control areas were created by this country.[14]

Another type of zone utilized in both times of tension and hostilities is the *cordon sanitaire*. Not universally recognized in international law because of its restrictive form, the establishment of a *cordon sanitaire* normally allows a nation to engage and destroy a potential enemy or unidentified aircraft or vessel without further notice.[15] Action such as this could be considered unreasonable in peacetime and has certain dubious characteristics in wartime.[16]

---

12. C. HYDE, INTERNATIONAL LAW, I 467 (1945).

13. Some examples are safety zones, protective areas, exclusive zones, identification zones, *cordons sanitaire*, self-defense areas, zones of responsibility, security and warning zones.

14. M. WHITEMAN, 4 DIGEST OF INTERNATIONAL LAW 389 (1963).

15. Gilliland, *Submarines and Targets: Suggestions for New Codified Rules of Submarine Warfare*, 73 GEO. L. J. 975, 991-996 (1984-85). This comment contains a lengthy exposition on the use of zones in modern naval warfare. Of special interest is the discussion of the "Total Exclusion Zone" established by the British during the Falkland Islands crisis.

16. Regan, *International Law and the Naval Commander*, 107 U.S. NAVAL INST. PROC. 51 (1981). This article discussed declaration of "moving" zones that accompany a naval task force. Such a concept is analogous to the establishment of zones around space objects. The author points out the drawbacks of such moving zones: first, they require the task force to broadcast its current and planned positions well in advance, something naval vessels are not disposed to do for obvious reasons; second, the zones would follow the task force, even in heavily trafficked areas, and therefore could greatly impede freedom of the seas and navigation in congested areas. The article proposes only the use of stationary zones around geographic areas.

Numerous warning and temporary security zones have been created to warn ships and aircraft that portions of the high seas were to be used for testing nuclear weapons, ballistic missiles, or naval gunnery exercises.[17] Between 1946 and 1954, the United States established warning areas of from 30,000 to 400,000 square miles in the Pacific. These zones are considered legal due to the negligible interference with normal shipping. When such zones are established, they do not prohibit the entry of ships and aircraft and thus do not constitute a blockade, but only warn of dangerous activity in the zone.

Both Iran and Iraq have declared such zones in the Persian Gulf. The United States issued a Notice to Mariners (NOTMAR)[18] and Notice to Airmen (NOTAM)[19] in the Gulf region that warned ships and aircraft not to approach closer than 5 nautical miles (nm) without permission. Those craft violating these warnings risk defensive measures.

International law recognizes safety zones as well as warning zones. The 1958 Convention on the Continental Shelf states that safety zones of up to 500 meters can be established around man-made installations on the

---

17. MCDOUGAL, *supra* note 2, at 298.

18. NOTMAR: U.S. naval forces operating in international waters within the Persian Gulf, Strait of Hormuz, the Gulf of Oman, and the Arabian Sea north of twenty degrees north are taking additional defensive precautions against terrorist threats. All surface and subsurface ships and craft are requested to avoid closing U.S. forces nearer than 5 nautical miles without previously identifying themselves. U.S. forces, especially when operating in confined waters, shall remain mindful of navigational considerations of ships and craft in their immediate vicinity. It is requested that radio contact with U.S. naval forces be maintained on channel 16, 121.5 mz VHF or 243.0 mz UHF when approaching within five nautical miles of U.S. naval forces. Surface and subsurface ships and craft that close U.S. naval forces within five nautical miles without making prior contact and/or whose intentions are unclear to such forces may be held at risk by U.S. defense measures.

These measures will apply when U.S. forces are engaged in transit passage through the Strait of Hormuz or when in innocent passage through territorial waters and when operating in such waters with the approval of the coastal state.

This notice is published solely to advise that measures in self-defense will be exercised by U.S. naval forces. The measures will be implemented in a manner that does not impede the freedom of any vessel of any nation or state.

19. NOTAM: U.S. naval forces in the Persian Gulf, Strait of Hormuz, Gulf of Oman, and Arabian Sea (north of 20 degrees north) are taking additional defensive precautions against terrorist threats. Aircraft at altitudes of less than 200 ft. at ground level which are not cleared for approach/departure to or from a regional airport are requested to avoid approaching closer than five nautical miles and to establish and maintain radio contact with U.S. naval forces on 121.5 megahertz (mz) VHF or 243.0 mz UHF. Aircraft which approach within five nautical miles at altitudes less than 2000 ft. at ground level whose intentions are unclear to U.S. naval forces may be held at risk by U.S. defensive measures.

continental shelf. A later comprehensive treaty incorporates the same concept of safety zones around artificial islands, research facilities and mining activities.[20] Lest the reader be left with the impression that this state of affairs arises from treaty provisions, the United States unilaterally declared safety zones around radar platforms as far as 100 miles away from our coast.[21]

A declared zone that has gained widespread acceptance in peacetime is the air defense identification zone (ADIZ). As a condition of entry into United States airspace, civil aircraft must pass through this coastal zone. The pilots are required to report their position not less than one hour and not more than two hours average direct cruising distance from the United States. In establishing the ADIZ, the United States is not claiming sovereignty over the high seas; the consequence of failing to comply with the rules for the ADIZ is the risk of interception by military fighter aircraft and a possible denial of entry into U.S. airspace.

In 1951, Canada promulgated a similar regulation in the interest of national security, the Canadian Air Defense Identification Zone (CADIZ). The CADIZ requirements are more severe than those of the U.S. ADIZ in that position reports must be made within the defense zone

---

20. United Nations Convention on the Law of the Sea, Article 60 states:

(4) The coastal State may, where necessary, establish reasonable safety zones around such artificial islands, installations and structures in which it may take appropriate measures to ensure the safety both of navigation and of the artificial islands, installations and structures.

(5) The breadth of the safety zones shall be determined by the coastal State, taking into account applicable international standards. Such zones shall be designed to ensure that they are reasonably related to the nature and function of the artificial islands, installations or structures, and shall not exceed a distance of 500 meters around them, measured from each point of their outer edge, except as authorized by generally accepted international standards or as recommended by the competent international organization. Due notice shall be given of the extent of safety zones.

(6) All ships must respect these safety zones and shall comply with generally accepted international standards regarding navigation in the vicinity of artificial islands, installations, structures and safety zones.

(7) Artificial islands, installations and structures and the safety zone around them may not be established where interference may be caused to the use of recognized sea lanes essential to international navigation.

(8) Artificial islands, installations and structures do not possess the status of islands. They have no territorial sea of their own, and their presence does not affect the delimitation of the territorial sea, the exclusive economic zone or the continental shelf.

United Nations Convention on the Law of the Sea, Dec. 10, 1982, art. 60, U.N. Pub. E. 83. V. 5 (1983) 21 I.L.M. 1261 (1982).

21. MCDOUGAL, *supra* note 2, at 298. These were "Texas towers," radar built on structures similar to oil drilling rigs.

whether or not the aircraft is bound for Canada.[22] In 1961, the French established a "zone of special responsibility" extending up to 80 miles off the coast of Algeria. The French did so in the belief that Algerian nationalists might use air routes to transport supplies and manpower.[23]

According to press reports, the French regulations required the filing of flight plans, identification of passengers flying within the assigned aerial corridor, and contact with ground identification stations. Apparently, the slightest infraction of any of these rules resulted in the interception of the offending aircraft by French military planes. [24]

Soviet Views

"In an effort to protect further their property in space, yet at the same time allow for the possibility of authorized visits, the Soviets have proposed the establishment of security zones *(zony bezopastnosti)* at some distance around their spaceships."[25] In the definitive work on Soviet space law positions, Zhukov and Kolosov state:

> That is why Soviet experts have raised the question of surrounding space objects with safety zones, within which the states on whose registry the objects are carried would exercise their sovereign rights of jurisdiction and control... The establishment of such zones could obviously not be tantamount to the appropriation of territory. International maritime law is familiar with the establishment of various zones of medical and customs control on the high seas adjacent to territorial waters.... Just as the establishment of such zones cannot be interpreted as appropriation of territory, so the establishment of safety zones around space objects cannot be seen as a sovereignty claim to the territory or space occupied by these zones. This is not just because such safety zones are temporary - they may be established for a sufficiently long period.... A territory under the sovereignty of some state differs essentially from any functional zones on territories in common use in that its status and forms of use are not subject to international

22. MCDOUGAL, *supra* note 2, at 308.

23. *Id.*

24. *Id.*, at 309.

25. Russell, *Military Activities in Outer Space: Soviet Legal Views*, 25 HARV. INT'L L. J. 155, 184 (1984).

> settlement, but are determined exclusively by the authority of the state exercising sovereignty over it.[26]

Other jurists are no less supportive of the concept of zones around space objects. *Vitaly D. Bordonov* predicts:

> The development of rapidly maneuverable space objects brings to the fore the problem of insuring the necessary conditions for the safe and effective functioning of space objects, especially those carrying a crew on board. For this purpose the state involved may establish around them zones of security within the bounds of which all the activities of space objects finding themselves in such zones must proceed in keeping with previously agreed stipulations. In the case of a threat to one space object from another stemming from a breach of rules of maneuvering in a zone of this kind, the state concerned has a right to take protective measures to remove this threat.[27]

The concerns expressed by Soviet writers vary; what does not is their expression of the need for and legality of exclusionary zones in space. Witness the comments of *B. G. Dudakov* arguing for a Convention to specify the distance limit around satellites to prevent "cosmic piracy."[28] Of great interest is this author's discussion of what constitutes this new form of piracy. He claims that:

> Even short-duration stationing (without any activities on the part of the pirate) in the vicinity of the satellite, which, as a rule, is equipped with rather sensitive scientific devices...may result in interference and substantially affect the satellite performance.[29]

The Soviet Union actively campaigned for years to have the U.S. Space Shuttle considered an ASAT weapon.[30] The vigor of these efforts has diminished as the Soviet Union comes closer to realizing an opera-

---

26. ZHUKOV & KOLOSOV, INTERNATIONAL SPACE LAW 64 (1984).

27. V. Bordonov, *Rights of States as Regards Space Objects,* PROC. 24TH COLLOQ. L. OUTER SPACE 90 (1981).

28. Dudakov, *On International Legal Status of Artificial Earth Satellites and the Zone Adjacent to Them,* PROC. 24TH COLLOQ. L. OUTER SPACE 97 (1981).

29. *Id.*, at 100.

30. Deudney, *Unlocking Space,* FOREIGN POL'Y (Winter 1983-84 at 101).

tional space plane of its own. The complaint of *Dudakov* and others is that the Shuttle's ability to pick up space objects in orbit foreshadowed "the prospective investigation of Soviet satellites and their destruction, if necessary."[31]

Implicit in the works of all Soviet international lawyers considering the issue is that the exclusionary zones are an inherent right of the state of registry. While advocating international agreements to accomplish this end, not one Russian lawyer has ever denied the possibility of a unilateral declaration of exclusionary zones.

While Soviet legal scholars have created the framework necessary to establish zones around space objects, criticism of the U.S. SDI program and such zones is reflected in technical writings of Soviet scientists. In their review of the vulnerability of SDI to Soviet countermeasures, a group of scientists ridiculed "sovereignty zones" because

> ...the size of the zone would be too small to protect a station against antisatellite weapons, say space mines. At the same time, these zones could not be extended indefinitely without disrupting normal peaceful activities in space.[32]

Absent from the quoted portion of the article and elsewhere in the book is any suggestion that the establishment of sovereignty zones would be violative of international law.

U. S. Views

No true consensus exists among Western authorities concerning space defense zones. At one end of the spectrum, analyses such as *Colin Gray's* represent the anti-zone positon:

> Unfortunately, there is no legal basis for an American assertion of sovereign prerogatives over space zones around its satellites. By way of analogy again, the U. S. would be asserting the right to sink, on suspicion only, any Soviet (or other) vessel that approached too closely, in unilateral U. S. judgment, to a U. S. aircraft carrier in time of peace or crisis. The obvious fact that the suspicious trailing object (candidate space mine) in space would be unmanned would render a U. S. decision to shoot somewhat less burdensome than in the case of a trawler at sea, but it is still virtually inconceivable that the U. S. would destroy Soviet spacecraft

31. Dudakov, *supra* note 28, at 98.

32. Y. VELIKHOV *et al.*, WEAPONS IN SPACE: THE DILEMMA OF SECURITY 46 (Moscow, Mir Press, 1986).

> in peacetime on suspicion. Not to mince words, such destruction, unsanctified by international law, would be an act of war. Preventive or preemptive self-defense has its strategic merits as a code of conduct, but it also invites anarchy. However, the potential U. S. (or Soviet - these arguments could cut both ways) dilemma cannot be ignored, even if there is no obvious plausible solution. A $3 million space mine could disable or destroy a $5 billion HEL battle station.[33]

Gray's analysis is flawed in a number of respects: As is amply demonstrated above and below, many of the world's most notable legal scholars in both the East and West have provided legal justification for keep-out zones; secondly, the destruction of a trespassing space object would not be "unsanctified" by international law and under most circumstances not amount to an act of war; and finally, preemptive self-defense hardly amounts to anarchy. Under all but the most unusual circumstances a *reasonable* application of force enhances stability.

Gray does concede that, given the realities of such a scenario, if the United States placed battle stations in orbit "that were absolutely critical to the U. S. theory of national security, it would have to bite the bullet-...and enforce an expansively defined 'Keep-Out Zone.'"[34]

The majority U. S. position might best be described as uncertain.[35] Most writers have a great deal of trouble reconciling keep-out zones and article II of the Outer Space Treaty which reads:

> Outer Space, including the moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means.[36]

This article is one of the foundations of the current legal regime for outer space. The original expression of this principle is found in United Nations General Assembly Resolution 1348, reflecting a desire of the General Assembly to avoid in outer space the rivalries caused by claims of sovereignty. The concept was repeated in Resolutions 1472,

---

33. C. GRAY, AMERICAN MILITARY SPACE POL'Y, 67 (Cambridge: ABT Books, 1982).

34. *Id.*, at 68.

35. DalBello, *supra* note 5, at 10.

36. Treaty on Principles Governing the Activities of States in the Exploration and Use of Outer Space, Including the Moon and Other Celestial Bodies January 27, 1967, art. II 18 U.S.T. 2410, T.I.A.S. 6347, 610 U.N.T.S. 205 (effective October 10, 1967) [Outer Space Treaty].

1721, and 1962. Article II is considered declaratory of customary international law, binding on all nations irrespective of whether they are party to the Outer Space Treaty or not.[37] This prohibition against national appropriation must be read in conjunction with the provisions of the entire Treaty, including article I, a portion of which states that

> Outer space, including the moon and other celestial bodies, shall be free for exploration and use by all States without discrimination of any kind, on a basis of equality and in accordance with international law, and there shall be free access to all areas of celestial bodies.[38]

Thus, appropriate non-aggressive uses of the space environment are permitted as long as no claim of exclusive use is based on sovereignty. Certain areas of the earth's surface are considered *res communis*, that is, the territory of no nation, available for all to use. However, historically the international community has recognized the right of jurisdictional competence, but not sovereignty over limited areas of what is recognized as *res communis*. This limited jurisdictional authority must meet the criteria of reasonableness under the circumstances, both with regard to the size of the zone created and the restrictions imposed on the use of the zone by other nations. *Professor McDougal* has eloquently set forth this standard and reminded us that reciprocity would protect against abuse. In elaborating on the test, he explained:

> For the detailed appraisal of the lawfulness of particular claims to exercise occasional exclusive competence in outer space, the most appropriate test would appear to be the traditional one of reasonableness. It is by this test that the authoritative decision-makers of the world community have consistently during modern times, with respect to many different sharable resources, decided between competing claims to inclusive use and authority, as well as upon the lawfulness of many different claims seeking to restrict that inclusivity for a variety of exclusive purposes. The burden of proving reasonableness should of course in each particular instance fall upon the state applying its authority in outer space to the spacecraft of other states. The reasonableness of the measures undertaken may be determined on the basis of a careful balancing of all the variable factors in context, with special emphasis upon the realities of the asserted threats. The multifactoral analysis

37. N. M. MATTE, SPACE ACTIVITIES AND EMERGING INTERNATIONAL LAW 275 (Montreal, CRASL, 1984).

38. Outer Space Treaty, art. I.

> we recommend for the appraisal of reasonableness of particular claims should, further, consider the relation between the degree of authority asserted and the degree of authority necessary to the achievement of the purpose, the area affected by such measures, especially its importance to other members of the international community and, generally, its significance for the conduct of peaceful exploration of space, the level of crisis in the earth-space arena at the time of assertion, the availability or the absence of alternatives that might achieve the same purpose but with less cost, the duration of interference resulting from the claim, and so forth.
>
> Any claim of such occasional competence by one state must, of course, carry the promise of reciprocity, in comparable circumstances, to all other states. The best protection of the general community against abuses in the exercise of exclusive competence may be found in the imperatives of clarifying a common interest, including both mutual tolerance and the realistic expectation of prompt retaliation against unreasonable demand. The requirements of "giving" for "taking," thus, impose necessary limits upon arbitrary and wanton assertions of authority.[39]

If such zones are created and they unduly interfere with space navigation, the zone created would be deemed unreasonable and therefore not legal under international law. If reasonableness is the correct test, and the author submits that *Professor McDougal* writing a quarter of a century ago was correct, zones around space objects need not be rejected *per se* based on the articles of the Outer Space Treaty. Applying well-recognized principles of international law developed for terrestrial application, such theories can be applied in the space environment.[40]

---

39. MCDOUGAL, *supra* note 2, at 293.

40. Professor DeSaussure has stated: "The Outer Space Treaty provides that outer space, including the moon and other celestial bodies, is not subject to national appropriation by claim of sovereignty, by means of use or occupation, or by any other means. Whether space stations are treated as instrumentalities or as extended territorial enclaves, this prohibits the creation of any territorial buffer zones around them. It may not, however, prohibit the creation of circumventing non-sovereign zones for safety or security. Space stations may eventually be regarded as real rather than personal property when they begin to shelter human settlements. They may then be classified as territory, floating space islands in constant orbit around the earth. The station's function, location, design, permanence, orbital path and habitability distinguish it from other space objects. It has been compared to a deepwater port, which is a fixed place on the high seas serving as a way station. These ports have a special status." DeSaussure, *The Impact of Manned Space Stations on the Law of Outer Space,* 21 SAN DIEGO L. REV. 985, 997 (1984) (footnotes omitted).

Precedent exists for declaring at least a temporary exclusive use of a *res communis* as described above when nations conduct naval maneuvers, satellite launches and recoveries, missile and nuclear bomb tests.[41]

A number of American policy analysts are consistent with the views of the Soviet jurists. One such example is the "Blau-Goure Doctrine" based on the principle that the exercise of the right of self-defense is part of national policy and international law, even in space.[42] These authors recognize the important political problems to be faced including the measured definition of zones, their purposes and their effective presentation publicly.[43]

Returning to the proposals of the OTA,[44] the rationale for the rules pertaining to keep-out zones fairly well summarize Western arguments in their favor:[45]

> - First, ASAT weapons such as nuclear interceptors would have to be kept at a range of several hundred kilometers from moderately hardened satellites in order to protect them;
>
> - Satellites in geostationary orbit are already so closely spaced that keep out zones large enough to protect them could not be established without displacing existing satellites and reducing the number of available "slots";
>
> - Third, a number of critical military missions could be performed by satellites in supersynchronous orbit, that is, above the geosynchronous orbit. In this region there would be adequate room to accommodate large keep-out zones;
>
> - Fourth, in the deep space area below geosynchronous orbit, several satellites normally occupy the same orbit, such as the highly elliptical, semi-synchronous "Molniya" orbits. Again, adequate room exists to establish keep-out zones of several hundred kilometer radius.

---

41. DalBello, *supra* note 5, at 10.

42. Blau & Goure, *supra* note 10, at 39. The Outer Space Treaty, art. III states that "State parties shall carry on activities in the exploration and use of outer space in accordance with international law, including the Charter of the United Nations...." *See infra* note 48 (discussion of the right to self-defense).

43. *Id.*, at 40.

44. *See supra* note 6.

45. *Supra* note 6, at 136.

- The proliferation of satellites in low-Earth orbit makes keep-out zones impractical. However, smaller zones could be established and different orbital planes could be required;

- Pre-announced close approaches would be permitted. These would be primarily for the purpose of inspection to determine if a satellite had a nuclear weapon on board.

In conclusion of its discussion on keep-out zones, the OTA recognized that any such regime will need careful tailoring to prevent such abuses as establishing so many zones to in effect create a space blockade.

Conclusion

While a number of conclusions can be drawn from the discussion above, certain obvious ones must be drawn:

- There may be no absolute legal impediment to the establishment of zones around any type of space object. The huge expanses of space and the speeds of spacecraft will allow and require substantial zonal dimensions. It is unreasonable to expect the crews of manned objects in space to allow the uncontrolled approach of other satellites. Strategic stability will be enhanced as the superpowers and their constellations of allies find a certain degree of security for their critical satellites.

- The establishment of zones would not seek to limit the production or deployment of space weapons; this would be left to other agreements. It would set up an information and clarification system whereby ambiguous acts in space could be explained. The Archduke Francis Ferdinand of World War III may very well be a critical Soviet or American satellite hit by a piece of space debris during a time of crisis.[46] Minimum distance requirements and the prohibition against close approaches are only an extension of well-tested terrestrial examples.[47]

46. Duedney, *supra* note 30, at 98.

47. For example, the US-Soviet Agreement on Prevention of Incidents at Sea, May 1972, which requires surveillance ships to "stay at a distance which avoids the risk of collision and...avoid executing maneuvers...endangering the ships under surveillance," Art. 3(4). *See* Stojiak, *Current Proposals for the Future Control of Outer Space Weaponization*, 10 ANNALS AIR & SPACE L. 453, 469 (1985), and S.M. Lynn-Jones, "A Quiet Success for Arms Control: Preventing Incidents at Sea," INT'L SECURITY 154-85 (1985).

- Based on the extensive expression of approval by Soviet jurists, the Soviet Union would have no legal soul-searching if it chose to implement zones around its space objects. While Western scholars debate the merits of space zones, our Eastern colleagues stand ready to establish them.

- From a military point of view, keep-out zones may not be the most efficient scenario to protect critical national security assets. Policing these areas may be close to impossible. Contrary to popular belief, NORAD does not constantly observe every satellite in orbit every minute of every day. This year, a Soviet satellite passed one of ours in geosynchronous orbit and it was not detected by NORAD.

- Military space planners are moving now to consider proliferation to be a reasonable way of ensuring survivability against ASAT's. An increase in the number of space objects will greatly complicate the task of keeping space defense zones under observation, as well as difficulty in enforcing transgressions in the zone. From a purely cost-effectiveness perspective, it may be more desirable to have a redundant system capable of sustaining losses with minimal mission impairment than trying to protect one extremely expensive, vital satellite that must be observed constantly and protected. This approach of deploying a hundred Volkswagons rather than one Rolls-Royce also makes sense when one considers the possibility of accidental destruction by collision with space debris or some catastrophic system failure.

- Notwithstanding all of the other military considerations, some thought must be given to the plausibility of the Soviet Union actually being able to accomplish a successful, simultaneous attack on every critical U. S. space asset. Anything less could put the Soviets at a disadvantage during a preemptive first strike scenario. Yet, the reality of the situation becomes mind-boggling when one considers that a potential aggressor would have to rely on largely untested weapons and untried tactics, with only one chance for success, under circumstances that would tax the most sophisticated command, control, and communications system, which the Soviets have so far failed to produce. A more practical approach to the protection of space assets may be derived from measures other than drawing a line in space and daring the other side to cross it.

- The defense of space-based assets would be consistent with the Charter of the United Nations. Article 51 of the Charter recognizes the inherent right of self-defense if an armed attack occurs. Even if one argued successfully that the placement of space-mines next to valuable satellites did not amount to an armed

attack, the U. S. position has always been that the pre-existing law with regard to self-defense survives unimpaired by any provision of the Charter. In the view of *Professor Oscar Schachter*, international law permits the use of legitimate self-defense in advance of actual attack.[48] He states that

> ...we must recognize that there may well be situations in which the imminence of an attack is so clear and the danger so great that defensive action is essential for self-preservation.[49]

The purpose of this paper has been to demonstrate certain legal and policy arguments that can be made in favor of keep-out zones should the necessity arise to establish them. No agency of the U. S. Government has advocated such a practice in the near future. The article began with a speculation on the part of SDIO on the efficacy of keep-out zones to protect against space mines. Such zones arguably could be established consistent with accepted principles of international law; however, no need has yet prompted such a move. For a variety of reasons, space mines and keep-out zones may be bad ideas whose time will never come. The categorical rejection of space zones for various purposes of safety, security, and traffic management, however, would be a grave mistake. If we truly believe that space can be colonized by earthkind, these concepts should be considered to promote safety, security, and stability.

48. Schachter, *The Right of States to Use Force*, 82 MICH. L. REV. 1620, 1634 (1984).

49. *Id.*